

WELL COMPLETION REPORT (SYNTHETIC)

OIL-LKW-011 - Lakwa-011

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000124
Field / District	Lakwa / Charaideo
Block	PEL-ASSAM-04
Basin	Assam-Arakan
Formation	Tipam (TIP)
Spud / Completion	2017-03-25 / 2017-06-04
Measured / TVD	2729 m / 2694 m
Trajectory	Directional; inc 26.8 deg; az 52.0 deg
Formation Interval	1291-2048 m
Lithology	Fine-medium sandstone with shale interbeds
Pore Pressure Reference	2573 psi
Fracture Gradient	14.41 ppg
Mud System	PHPA Water-Based Mud
NWIS Well Risk	High (57.3/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	682	Surface
9 5/8 in casing	1582	Intermediate
7 in liner/casing	2401	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2017-04-04 538 m	Mud Loss	Critical	Flow-out dropped below flow-in with pit loss. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Track flow balance and pit volume trend before losses become severe.
2017-04-26 1359 m	NPT	Critical	ROP dropped to zero during equipment unavailability. Mitigation: Classify NPT cause consistently for offset-well learning. Lesson: Separate geological risk from equipment-driven NPT in analytics.
2017-05-14 1996 m	Fishing Operation	Critical	Downhole tool remained unrecovered after freeing attempts. Mitigation: Review BHA failure mode before resuming drilling. Lesson: Early stuck-pipe escalation may reduce probability of fishing.
2017-04-27 1379 m	Mud Loss	High	Partial returns observed at shale shaker. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Track flow balance and pit volume trend before losses become severe.
2017-05-12 1937 m	Kick	High	Unexpected background gas rise preceded influx. Mitigation: Shut in, verify influx and circulate using approved well-control procedure. Lesson: Maintain mud-weight window using offset pressure evidence.
2017-05-21 2249 m	Pressure Spike	High	Short-duration high-pressure excursion was recorded. Mitigation: Use pressure trend thresholds referenced to depth and flow rate. Lesson: Use depth-matched offset hydraulic baseline for early warning.
2017-05-25 2384 m	Mud Loss	High	Intermittent circulation losses developed while drilling ahead. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.

Completion Summary

The synthetic well reached 2729 m MD in the Tipam interval. The NWIS knowledge record links 17 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.